《计算机网络》作业

第十一次作业：(英文版教材第六章 8，10，11，7-4，1-20，1-12，18，21)

8. (6.3.1 Desirable Bandwidth Allocation)In Figure 6-20, suppose a new flow E is added that takes a path from R1 to R2 to R6. How does the max-min bandwidth allocation change for the five flows?

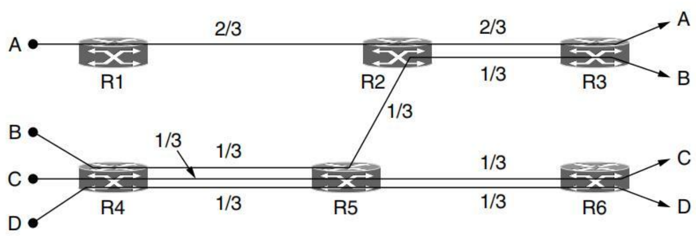

Figure 6-20. Max-min bandwidth allocation for four flows.

10. (6.3.2 Regulating the Sending Rate)Some other policies for fairness in congestion control are Additive Increase Additive Decrease (AIAD), Multiplicative Increase Additive Decrease (MIAD), and Multiplicative Increase Multiplicative Decrease (MIMD). Discuss these three policies in terms of convergence and stability.

11. (6.4.1 Introduction to UDP)Why does UDP exist? Would it not have been enough to just let user processes send raw IP packets?

7-4.(6.4.1 Introduction to UDP)In addition to being subject to loss, UDP packets have a maximum length, potentially as low as 576 bytes. What happens when a DNS name to be looked up exceeds this length? Can it be sent in two packets?

1-20. (6.4.1 Introduction to UDP & 6.5.1 Introduction to TCP)What is the similarity of UDP and TCP? What is the main difference between them?

1-12.(6.5.1&6.5.2-Features of TCP)Two networks each provide reliable connection-oriented service. One of them offers a reliable byte stream and the other offers a reliable message stream. Are these identical? If so, why is the distinction made? If not, give an example of how they differ.

18. (6.5.1-6.5.3 TCP Features)Datagram fragmentation and reassembly are handled by IP and are invisible to TCP. Does this mean that TCP does not have to worry about data arriving in the wrong order?

21. (6.5.1-6.5.3 TCP Features)A process on host 1 has bee6n assigned port p, and a process on host 2 has been assigned port q. Is it possible for there to be two or more TCP connections between these two ports at the same time?